

Alzikrayat Project Report

1. Project Overview

Alzikrayat is a photo sharing web application built as a course project for the Advanced Web Technologies course at Sudan University of Science and Technology (SUST).

The application allows users to register, log in, upload photos, browse a gallery, comment, like, and tag other users.

The project is built from scratch using PHP without a backend framework, following the course requirements for a handwritten MVC architecture and a manual routing system.

2. Architecture & Project Structure

The project follows two complementary architectures:

3-Tier Architecture

- **Presentation:** views/, public/
- **Application:** controllers/, core/
- **Data:** models/, config/

MVC Pattern

- **Model:** Handles database operations and SQL queries.
- **View:** Contains the HTML templates and user interface.
- **Controller:** Handles requests and connects the models with the views.

Custom Router

The Router in core/Router.php maps URLs to controllers using Regular Expressions.

For example, /photo/{id} becomes the pattern:

```
| #^/photo/([^/+]$)#
```

and calls:

```
| PhotoController::show($id)
```

Project Structure

```
| alzikrayat/  
| └── config/      # Database connection  
| └── core/        # Router, Model, Controller  
| └── controllers/ # Auth, Home, Photo, Comment, Like  
| └── models/      # User, Photo, Comment, Like, Tag  
| └── views/       # All HTML templates  
| └── public/      # Entry point, CSS, JS, uploads
```

3. Database Design

The database consists of five tables with primary keys, foreign keys, unique constraints, and cascade deletion.

1. users

```
id          INT, PRIMARY KEY, AUTO_INCREMENT
first_name  VARCHAR(50), NOT NULL
last_name   VARCHAR(50), NOT NULL
email       VARCHAR(100), NOT NULL, UNIQUE
password    VARCHAR(255), NOT NULL
location    VARCHAR(100), NULL
description TEXT, NULL
occupation  VARCHAR(100), NULL
```

The users table stores the registered users and their account information.

2. photos

```
id          INT, PRIMARY KEY, AUTO_INCREMENT
user_id     INT, FOREIGN KEY, NOT NULL
file_name   VARCHAR(255), NOT NULL
title       VARCHAR(200), NOT NULL
description  TEXT, NULL
date_time   TIMESTAMP, NOT NULL
```

The photos table stores uploaded photos and their related information.

3. comments

```
id          INT, PRIMARY KEY, AUTO_INCREMENT
photo_id    INT, FOREIGN KEY, NOT NULL
user_id     INT, FOREIGN KEY, NOT NULL
comment     TEXT, NOT NULL
date_time   TIMESTAMP, NOT NULL
```

The comments table stores comments made by users on photos.

4. likes

```
user_id     INT, FOREIGN KEY, NOT NULL
photo_id    INT, FOREIGN KEY, NOT NULL
created_at  TIMESTAMP, NOT NULL
```

The composite primary key is: PRIMARY KEY (user_id, photo_id)

This prevents the same user from liking the same photo more than once.

5. tags

```
id          INT, PRIMARY KEY, AUTO_INCREMENT
photo_id    INT, FOREIGN KEY, NOT NULL
user_id     INT, FOREIGN KEY, NOT NULL
created_at  TIMESTAMP, NOT NULL
```

The table also uses: UNIQUE KEY (photo_id, user_id)

to prevent the same user from being tagged more than once in the same photo.

Referential Integrity

Foreign keys use ON DELETE CASCADE.

When a user or photo is deleted, the related photos, comments, likes, and tags are removed automatically according to their relationships.

4. Main Features

1. User Authentication

The application provides user registration, login, and logout.

Passwords are stored using Bcrypt password hashing. Sessions are used to keep users logged in, and a last-login cookie stores the last login timestamp for 7 days.

2. Home Page

The home page contains dynamic statistics, recent photos, navigation, and an About page.

3. Photo Management

Users can upload photos with validation, browse the photo gallery, view photo details, and delete their own photos.

4. Comments

Logged-in users can add comments to photos.

Each comment is connected to the logged-in user and the selected photo and includes a timestamp.

5. Validation

The application uses three levels of validation:

- HTML5 validation
- JavaScript client-side validation
- PHP server-side validation

These checks help validate user input before it is stored in the database.

6. Security

The application uses:

- PDO prepared statements
- htmlspecialchars()
- Password hashing
- Ownership verification

- File upload validation

These measures help protect the application against common security problems such as SQL injection and XSS.

7. Responsive Design

The interface uses Bootstrap 5 together with custom CSS to support different screen sizes.

8. Custom 404 Page

A custom-designed 404 page is displayed when a requested page is not found.

5. Novelty / Advanced Features

1. Photo Likes System

Users can like or unlike photos.

A composite primary key prevents duplicate likes.

The like count and heart animation appear on each photo.

2. Photo Tagging System

Users can tag other registered users in photos.

Tags are stored in a dedicated database table and displayed on the photo detail page.

3. Custom Grid Toggle

The gallery can switch between:

- 3-column view
- 4-column view
- List view
- Slider view

The selected view is saved in localStorage.

4. Last-Login Cookie

A 7-day cookie stores the last successful login timestamp and displays it on the login page.

6. Conclusion

Alzikrayat demonstrates a full-stack web application built using a handwritten MVC structure and a manual regex-based router.

The project applies Bcrypt password hashing, PDO prepared statements, XSS protection, ownership checks, input validation, and file upload validation.

Key Achievements

- Custom regex-based Router
- Manual MVC separation
- 3-Tier architecture
- Secure authentication
- Client-side and server-side validation
- SQL injection and XSS protection
- Foreign key cascades
- Photo upload and deletion

- Comments system
- Likes system
- Tagging system
- Custom gallery display styles
- Responsive design with Bootstrap 5